

Class 25

AI and the Institutions of Progress

Part IV — Governing Capitalism's Consequences

The American Economy — draft course materials (proposal under discussion)

The point of today's class

The next American growth question is not whether AI raises GDP. It is whether institutions can convert rapid idea-driven progress into broad-based welfare — while managing attention, physical-world bottlenecks, inequality, and catastrophic risk.

Why it matters

- ▶ Ask yourself: what will be *scarce* in 2050?
 - ▶ Probably not information, manufactured goods, or routine expertise.
 - ▶ Maybe: attention, trust, clean firm power, healthy life-years, housing near opportunity, focus, and meaning.
- ▶ Welfare changes when scarcity changes: a society can grow materially richer while struggling with attention, housing, health, and trust.
- ▶ This is the last class — the course turning back on itself. America built a growth machine out of institutions. Can those institutions handle what comes next?

Today

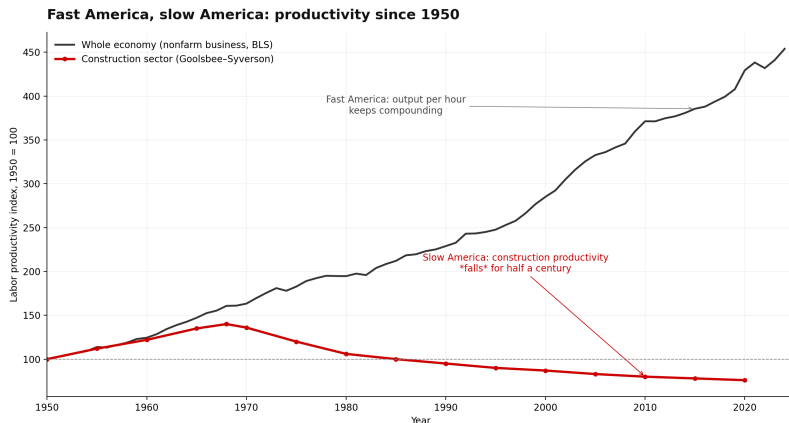
1. The course in one slide
2. Fast America, slow America
3. AI as a general-purpose technology
4. Cheap expertise — and who gains
5. The Baumol bottleneck
6. Speed vs. risk
7. The closing argument

The course in one slide

What we learned	The finale question
Growth transformed human life	Is AI the next growth wave?
Growth runs on ideas, firms, scale	AI needs chips, energy, institutions
Talent must be allowed to participate	Who gets access to AI?
Markets create externalities, crises, power	What new risks need governing?

- ▶ The course thesis, one more time: America's economy is a growth machine built from institutions.
- ▶ The finale question: can those institutions adapt quickly enough?

Fast America, slow America: productivity since 1950



Whole economy: BLS nonfarm business output per hour via FRED (OPHNFB), annual averages. Construction: approximate values from Goolsbee and Syverson (2023), 'The Strange and Awful Path of Productivity in the U.S. Construction Sector,' Figure 1 (teaching reconstruction).

Source: Whole economy: BLS nonfarm business output per hour via FRED (OPHNFB). Construction: approximate values from Goolsbee and Syverson (2023), Figure 1 (teaching reconstruction).

Fast America, slow America

- ▶ Operation Warp Speed, May 2020: a goal of 300 million doses, first doses by January 2021 — vaccines for a new virus in months.
- ▶ Construction productivity has stagnated or fallen for five decades (Goolsbee–Syverson); residential construction is back near its 1930s level.

Fast America	Slow America
AI models	Housing construction
mRNA vaccines	Transmission lines
Chips and software	Transit and nuclear power
Venture-backed scaling	Permitting and land assembly

- ▶ America invents in the world of bits faster than it can build in the world of atoms. Why? Not simply “too much regulation” — veto points, environmental review, litigation, fragmented authority, procurement rules.

AI as a general-purpose technology

GPT	Core invention	Complementary investments
Steam	Engine	Mines, railways, factories
Electricity	Motors, generation	Grids, factory redesign
Computers	Chips, software	Skills, databases, management
Internet	Networks, protocols	Broadband, platforms, logistics
AI	Models, chips, data	Compute, energy, workflows, trust, rules

- ▶ A GPT is not just an invention — it is a reorganization of the economy around an invention.
- ▶ Electricity took decades because factories had to be redesigned; computers took decades because firms had to reorganize information. AI may be similar.

Global corporate AI investment, 2014–2025

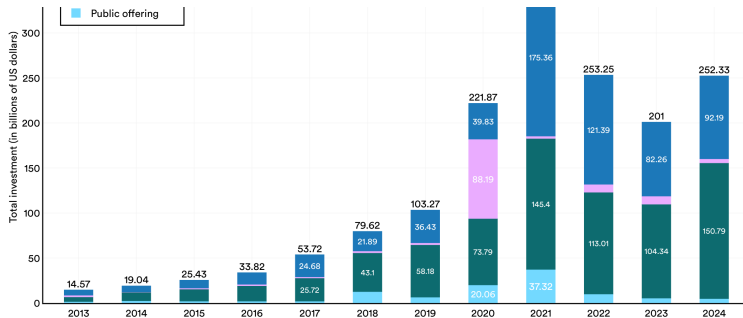


Figure 4.3.1

Source: Stanford Institute for Human-Centered AI, AI Index Report 2026. U.S. private AI investment \$285.9B in 2025.

Investment is not productivity

- ▶ The surge is real: U.S. private AI investment reached \$285.9 billion in 2025 — more than 23 times China's reported total — with 1,953 newly funded U.S. AI companies.
- ▶ Does investment prove productivity will rise? No — it shows expectations and a race to build capacity.
- ▶ Every AI input needs a complement:
 - ▶ chips need electricity and cooling
 - ▶ models need useful workflows
 - ▶ adoption needs trust, evaluation, and retraining
 - ▶ talent needs immigration and education
- ▶ AI is not a productivity revolution until ordinary firms use it to change how they actually produce.

Cheap expertise — and who gains

- ▶ The better frame is not “robots take jobs” but **cheap expertise**: tutoring, coding, drafting, diagnostic support, legal access, research assistance.
- ▶ If the Industrial Revolution made muscle cheap, AI may make some kinds of expertise cheap — though cheap is not the same as correct, trusted, or equally distributed.
- ▶ AI can *replace* tasks (routine writing, basic coding) or *augment* workers (better drafting, faster development, decision support) — the mix determines who gains.
- ▶ Lost Einsteins callback: cheap expertise could democratize invention and opportunity — or gains could concentrate behind compute, data, platforms, and elite schools.
- ▶ Productivity creates the *possibility* of higher wages; it does not distribute itself.

The Baumol bottleneck

Fast productivity sectors	Slow productivity sectors
Semiconductors, software, AI	Construction and housing
Digital media	Childcare and elder care
Biotech discovery	Clinical care delivery
Cloud computing	K–12 education

- ▶ As the fast sectors get cheap, the slow sectors dominate family budgets and constrain welfare.
- ▶ If AI makes software and knowledge nearly free but housing, health care, childcare, and energy stay expensive, the gains are bottlenecked.
- ▶ The future may be limited not by invention, but by the slowest institutions around the invention.

Speed vs. risk

Domain	Need for speed	Need for caution
AI	Productivity, competition	Safety, bias, misinformation
Biotech	Lives saved, pandemics	Biosecurity, ethics, trust
Housing	Affordability, mobility	Local externalities
Transmission	Decarbonization	Land use, local voice
Nuclear	Clean firm power	Safety, waste

- ▶ The same rules that protect (clean air, safety, community voice, due process) can block (housing, transmission, clean energy, urgent public goods).
- ▶ Good institutions are not simply fast or cautious — they are fast where delay is costly and cautious where mistakes are catastrophic.

Discussion

Should America move fast or slow? Pick one: AI tutors in schools, AI diagnosis, gene editing, housing permits, transmission lines, nuclear power. What is gained by speed, what is risked by error, who benefits from delay — and which institution should decide?

The closing argument

- ▶ We began with the hockey stick: for most of history, ordinary people were poor, sick, and short-lived. Then growth happened.
- ▶ But growth was never just machines and money. It was ideas, markets, law, infrastructure, schools, firms, immigration, finance, public goods, competition — and the repeated correction of capitalism's failures.
- ▶ AI and biotech may be astonishing. Whether they become broad-based progress depends on institutions: moving fast where speed saves lives, moving carefully where mistakes are catastrophic, and measuring prosperity by more than GDP.

Growth creates possibility. Institutions decide whether it becomes prosperity. Whether America's institutions keep adapting — that question now belongs to you.

Read before next class

There is no next class — these are readings for the road.

- ▶ Brynjolfsson, Rock, and Syverson, “Artificial Intelligence and the Modern Productivity Paradox” — why GPTs take time.
- ▶ Goolsbee and Syverson, “The Strange and Awful Path of Productivity in the U.S. Construction Sector.”